

STRATEGIC ALLIANCE

Inspironics AIoT,

"Measure. Connect. Understand. Predict. Optimize. Monetize."

Enmazz

Industrial Measurement · Data Acquisition ·
Power Quality · Sustainability Consulting ·
Field Instrumentation

Inspironics

AIoT Platforms · Digital Twin · ESG
Intelligence · Agentic AI · Hospitality
Technology · Smart Building Automation

Combined

Connected Assets → Digital Twins → AI
Intelligence → Autonomous Operations →
Sustainability Outcomes → Enterprise
Value

Industry 5.0 · Multi-Vertical · Open Platform · ESG Native · Agentic AI · Multi-Cloud

E x e c u t i v e

Position Inspironics and Enmaz as a unified **Industry 5.0 ecosystem** — combining hardware measurement excellence with AI-powered intelligence platforms to deliver end-to-end operational intelligence across every enterprise vertical.

E n m a z :

- Industrial Measurement & Data Acquisition
- Power Quality & Energy Monitoring
- Sustainability Consulting
- Field Instrumentation

I n s p i r o n i c s :

- AIoT Platforms & Digital Twin
- ESG Intelligence & Reporting
- Agentic AI & Autonomous Operations
- Hospitality Technology & Smart Building Automation

C o n n e c t e d A s s e t s

Sensor-rich devices
linked across
operations



A I

Predictive insights
and anomaly
detection



D i g i t a l

Real-time virtual
models of physical
systems



A u t o n o m o u s

Automated control
and decisioning loops

The combined entity delivers a seamless journey from raw physical-world data to autonomous enterprise decisions — a capability no single incumbent provides out of the box.

A l l i a n c e

The Inspironics–Enmaz alliance is architected around a clear division of excellence that, when unified, creates a category-defining Enterprise Operational Intelligence platform.

E n m a z

Measurement Excellence

Deep hardware expertise in sensors, data loggers, power quality, spectrum analysis, and field instrumentation — the trusted physical-world sensing layer.

I n s p i r o n i c s

Intelligence Excellence

AI-native platforms, digital twin engines, ESG compliance frameworks, agentic AI workflows, and multi-vertical SaaS — the intelligence and monetization layer.

T o g e t h e r

Enterprise Operational Intelligence

A unified ecosystem where every physical measurement becomes an intelligent decision — closing the loop from sensor to boardroom with full ESG accountability.

- The alliance creates a **whole greater than the sum of its parts**: Enmaz provides ground-truth data fidelity; Inspironics transforms that data into autonomous, monetizable intelligence.

C o m p l e t e

D e v i c e s

Enmaz Hardware:

- Data Logger · Spectrum Analyzer
- IoT Data Acquisition Unit
- IoT Vibration Sensor · IoT Air Quality Sensor
- IoT Protocol Converter
- Energy Monitoring Devices
- Power Quality Monitoring

Inspirionics Devices:

- Kleio IntelliHub™ · Mikos AirQ™
- Amenos™ · Afi™ · Kinisi™

P l a t f o r m s

~~C K~~-C a l e i d o
K o m b o s

Industrial IoT Platform

~~C E~~-C i e l o™

ESG + Digital Twin + AI
Platform

~~C X~~-C a l e i d o™

Hospitality Platform

~~C D~~-C a l e i d o™

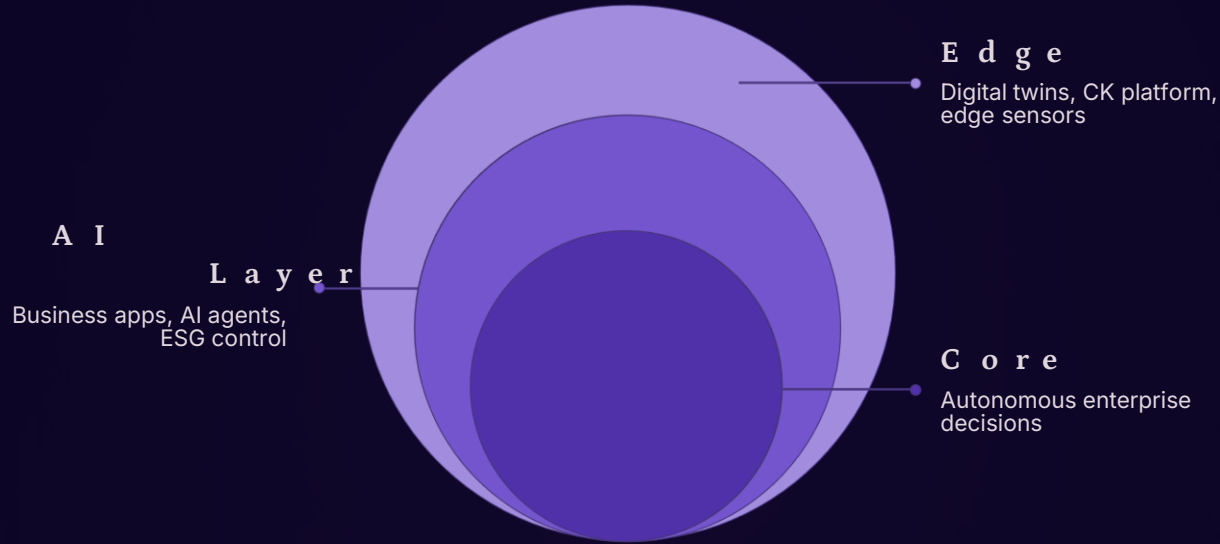
Occupant Experience
Platform

~~C M~~-C a l e i d o™

Engagement & Wellness Platform

I n d u s t r y

The full-stack architecture spans nine integrated layers — from raw physical assets to fully autonomous enterprise decisions. Every layer is independently valuable; together they form an unbreakable intelligence continuum.



P e r c e p t i o n

Physical Assets · Edge Sensors · Edge Intelligence

I n t e g r a t i o n

CK Platform · Digital Twin Layer

I n t e l l i g e n c e

AI Agent Layer · CE ESG Layer

A c t i o n

Business Applications · Autonomous Decisions

Device

The Inspironics device family provides comprehensive sensing coverage across every critical building and facility parameter. Each device is purpose-engineered and protocol-agnostic, connecting seamlessly through the Kleio IntelliHub™ gateway.



Environmental™

PM1, PM2.5, PM10, CO₂, VOC, Noise, Lux, Temperature, Humidity — full indoor environmental quality monitoring in a single device.



Comfort™

HVAC, FCU, Heat Pumps, VRF — intelligent climate control and comfort actuation across all room types.



Access™

RFID, BLE, Mobile Access, Visitor Management — unified access intelligence for people and spaces.



Occupancy™

Presence, Motion, Utilization — real-time space occupancy data feeding digital twin and optimization engines.



Gateway™

MQTT, BACnet, Modbus, REST, OTA — the universal edge gateway bridging physical devices to the CK Platform.

D i g i t a l

The Inspironics digital twin framework creates five distinct but interconnected virtual representations of the enterprise — enabling simulation, prediction, and autonomous optimization at every level of the built and operational environment.



B u i l d i n g

Hotels, Hospitals, Commercial Buildings — complete structural and systems virtualization for operational intelligence and scenario modeling.



A s s e t

Motors, Pumps, HVAC, Chillers, Boilers — asset-level digital replicas enabling predictive maintenance and lifecycle optimization.



U t i l i t y

Energy, Water, Waste, Carbon — real-time utility consumption models powering ESG reporting and net-zero planning.



O c c u p a n c y

People, Guests, Employees, Visitors — dynamic occupancy modeling for space optimization and experience personalization.



P r o c e s s

Operations, Maintenance, Sustainability — end-to-end process virtualization enabling continuous improvement and autonomous workflow execution.

M o d e r n

The combined platform is built on a battle-tested, cloud-native, AI-first technology stack — purpose-designed for the demands of Industry 5.0 deployments at enterprise scale.



A I

GenAI · LLM · Agentic AI · Copilots — intelligent reasoning and autonomous decision-making at every platform layer.



D a t a

Lakehouse · Data Fabric · Data Mesh — unified, governed, real-time data architecture for enterprise-scale analytics.



C l o u d

AWS · Azure · Google — multi-cloud deployment with no vendor lock-in, supporting hybrid and on-premise scenarios.



E d g e

TinyML · Edge AI · Federated Learning — intelligence at the point of data generation, reducing latency and bandwidth.



V i s u a l i z a t i o n

Digital Twin · AR · VR · Spatial Analytics — immersive, actionable representations of operational reality.

O p e n

The platform embraces an open-sensor philosophy — integrating best-in-class open source and commercial sensor hardware across every environmental domain, dramatically reducing deployment cost and accelerating adoption.



A i r

- AirGradient · SenseCAP
- PMS5003 · SCD41



E n e r g y

- OpenEnergyMonitor · PZEM
- Eastron



W a t e r

- Atlas Scientific
- DFRobot



A g r i c u l t u r e

- Soil Moisture · NPK
- EC · Weather



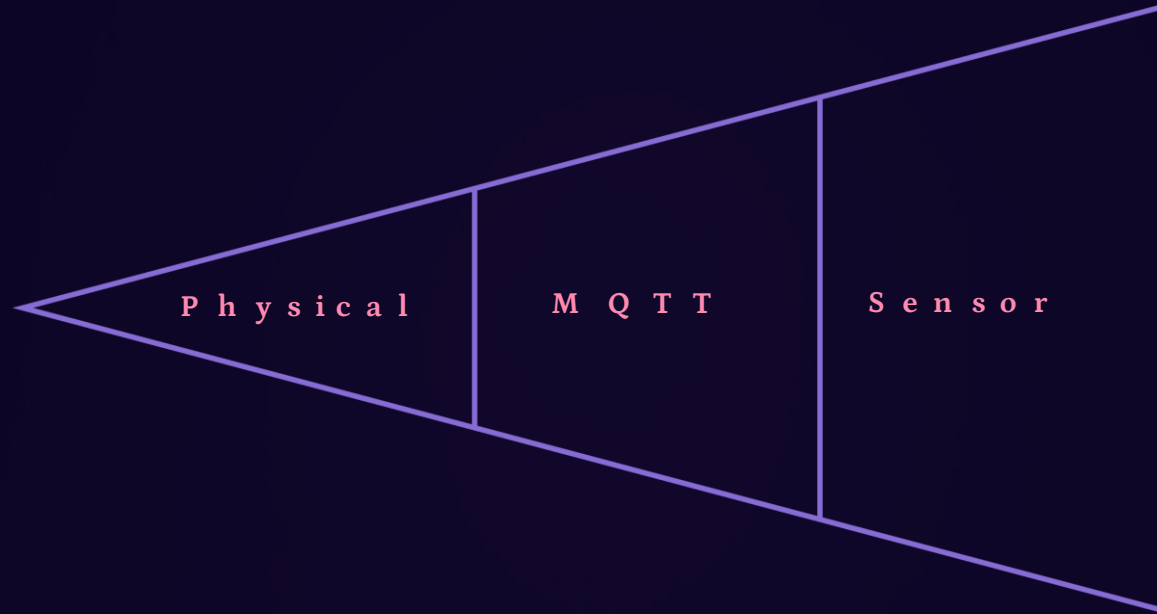
V i b r a t i o n

- MEMS Accelerometers
- Industrial Vibration Sensors

Open sensor compatibility lowers the total cost of deployment, enables brownfield retrofits, and expands the addressable market to cost-sensitive segments including SME industrials, agriculture, and public sector.

E n t e r p r i s e

Every operational insight begins with a physical event. The enterprise data traversal pipeline ensures that raw sensor telemetry is transformed — with zero data loss — into autonomous intelligent actions through a ten-stage processing architecture.



I n g e s t

- **Physical World:** Sensors, actuators, meters
- **Sensor Telemetry:** Structured & unstructured streams
- **MQTT Broker:** Lightweight, high-throughput pub/sub
- **Edge Processing:** Local filtering, anomaly detection

S t o r e ,

- **Time Series DB:** High-frequency operational data
- **Data Lake:** Historical, contextual, enriched data
- **Digital Twin:** Real-time virtual representation
- **ML Models → AI Agents → Actions:** Predict, decide, execute

Enterprise

The platform is anchored by a rigorous, hierarchical enterprise data model that provides consistent, governed data structures across every deployment — enabling cross-site benchmarking, AI training, and ESG reporting at scale.

Master

The organizational spine — every data point is contextualized within this structure:

- Organization → Region → Customer
- Site → Building → Floor → Zone
- Asset → Sensor → Gateway → User

Transaction

Operational events flowing through the master hierarchy:

Operational

Telemetry · Alerts · Events · Work Orders

Sustainability

Energy · Water · Carbon · Occupancy

Maintenance

Maintenance Records · Asset Lifecycle Data

- ❏ A unified master data model is the foundation for reliable AI training, accurate ESG accounting, and defensible audit trails — a key differentiator in regulated industries such as healthcare, finance, and public sector.

A I

The Agentic AI layer deploys six specialized autonomous agents — each purpose-built for a distinct operational domain. These agents operate continuously, consuming digital twin telemetry and enterprise data to deliver real-time decisions without human intervention.



O p e r a t i o n s

Energy Optimization — continuously tunes HVAC, lighting, and utility systems to minimize consumption while maintaining occupant comfort.



E S G

Carbon Compliance — monitors emissions against regulatory frameworks in real time, generating automated compliance evidence.



E x e c u t i v e

Strategic Insights — synthesizes cross-domain intelligence into boardroom-ready narratives, KPIs, and scenario analysis.



M a i n t e n a n c e

Predictive Maintenance — detects asset degradation patterns before failure, scheduling interventions at optimal cost-benefit windows.



O c c u p a n c y

Space Optimization — dynamically reallocates space resources based on real-time utilization data to maximize efficiency.



S u s t a i n a b i l i t y

Net Zero Planning — models decarbonization pathways, tracks progress against net-zero commitments, and recommends capital allocation.

H o s p i t a l i t y

The hospitality vertical benefits from the deepest platform integration — combining guest experience, building intelligence, and sustainability compliance into a single connected ecosystem powered by CX, CD, and CE platforms.

~~C X~~-C a l e i d o™

Guest Experience Layer

- Guest Journey Personalization
- Digital Concierge & Service Automation
- Intelligent Room Automation

~~C D~~-C a l e i d o™

Occupant Intelligence Layer

- Thermal Comfort Optimization
- Real-Time Occupancy Sensing
- Smart Access & Security

~~C E~~-C i e l o™

Sustainability Intelligence Layer

- Energy Consumption Monitoring
- Water Usage Analytics
- Carbon Footprint & ESG Reporting

- ✔ Hospitality operators leveraging the full CX + CD + CE stack can target **15–30% energy savings**, significant guest satisfaction uplift, and automated ESG compliance reporting across all properties.

H e a l t h c a r e

Hospitals and healthcare facilities represent one of the most demanding and high-value deployment environments — combining rigorous compliance requirements, 24/7 operational criticality, and complex energy management challenges.

P a t i e n t

Continuous monitoring of temperature, humidity, air quality, and lighting across patient rooms and clinical areas — automated to clinical comfort standards without manual intervention.

I n d o o r

Real-time IAQ monitoring including CO₂, VOC, PM2.5, and pathogen-risk indicators — critical for infection control and regulatory compliance in clinical environments.

E n e r g y

Hospital-grade energy optimization across HVAC, lighting, and medical support systems — targeting significant cost reduction without compromising clinical operations.

M e d i c a l

IoT-enabled asset tracking and performance monitoring for medical equipment — reducing downtime, extending asset life, and improving clinical workflow reliability.

C o m p l i a n c e

Automated generation of compliance evidence for JCIA, ISO, and local healthcare regulatory frameworks — dramatically reducing audit preparation time and risk.

D i g i t a l

Full building and systems digital twin enabling scenario simulation, capacity planning, energy modeling, and predictive maintenance across the entire hospital estate.

M a n u f a c t u r i n g

The manufacturing vertical demands the highest data fidelity and lowest latency — from shop-floor vibration telemetry to enterprise-level OEE dashboards. The combined Enmaz + Inspironics stack delivers the full spectrum.

O p e r a t i o n a l

- **Vibration Monitoring:** MEMS-based continuous vibration analysis on motors, pumps, and rotating equipment
- **Predictive Maintenance:** AI-driven failure prediction with days-to-failure forecasting and automated work order generation
- **OEE (Overall Equipment Effectiveness):** Real-time availability, performance, and quality metrics across all production lines

S u s t a i n a b i l i t y

- **Energy Analytics:** Per-machine, per-line, and factory-level energy consumption with waste identification and optimization recommendations
- **Digital Factory Twin:** A complete virtual replica of the factory floor enabling simulation, what-if analysis, and autonomous optimization
- **Asset Reliability:** Full asset lifecycle intelligence — from commissioning to decommissioning — with reliability-centered maintenance strategies

 Manufacturing clients typically achieve **20–40% reduction in unplanned maintenance costs** and **10–30% extension in asset operational life** within 12–18 months of full platform deployment.

D a t a

Data centers are among the most energy-intensive facilities in the modern economy — and among the most receptive to AI-driven optimization. The platform delivers measurable PUE improvement and operational resilience from day one.



R a c k

Per-rack power, temperature, and airflow monitoring with anomaly detection — eliminating thermal hotspots and capacity blind spots.



C o o l i n g

AI-driven cooling control that dynamically adjusts CRAC, CRAH, and containment systems based on real-time thermal load data.



E n e r g y

Granular energy visibility across IT load, cooling, lighting, and UPS — enabling precise identification of efficiency opportunities.



P U E

Continuous PUE monitoring and optimization targeting industry-leading efficiency ratios — directly impacting OpEx and ESG reporting.



D i g i t a l

A live digital twin of the entire data center enabling capacity planning, failure scenario simulation, and autonomous operational optimization.

S m a r t

Precision agriculture is the next frontier for AIoT — combining open-source sensor ecosystems with AI analytics to optimize yield, minimize water and input waste, and deliver farm-level ESG accountability.



W e a t h e r

Hyperlocal weather station networks providing real-time microclimate data — enabling precision irrigation scheduling, frost alerts, and crop protection decisions.



S o i l

Continuous soil moisture, NPK, EC, and pH monitoring — driving precision fertilization and irrigation that maximizes yield while minimizing input costs.



W a t e r

AI-driven irrigation control that reduces water consumption by up to 40% through demand-based scheduling calibrated to soil and weather data.



C a r b o n

Farm-level carbon accounting and Scope 3 emissions tracking — enabling agricultural enterprises to meet ESG commitments and access carbon credit markets.



C r o p

ML-powered crop health monitoring, yield prediction, and anomaly detection — integrating satellite imagery, sensor data, and agronomic models.

U t i l i t y

Cities and utilities represent the largest addressable market in the ecosystem — with city digital twins becoming the standard operating model for urban management. The platform's open architecture makes it uniquely deployable at city scale.

G r i d

Real-time electricity grid monitoring, load balancing intelligence, and renewable energy integration management at utility scale.

S m a r t

Adaptive lighting control based on occupancy, time, and weather — targeting 30–50% energy reduction in municipal lighting budgets.

W a t e r

Distribution network monitoring, leak detection, quality sensing, and pressure optimization across municipal water infrastructure.

W a s t e

Smart waste collection optimization — fill-level sensing, dynamic route planning, and contamination monitoring for cleaner, lower-cost urban waste management.

U r b a n

Dense city-wide air quality sensor networks providing hyperlocal pollution mapping to support public health decisions and environmental compliance.

C i t y

Comprehensive city-scale digital twin integrating all urban systems — enabling simulation, crisis response, and long-term urban planning at unprecedented fidelity.

ESG

ESG compliance is no longer optional — it is a regulatory, investor, and market requirement across every major economy. The CE — Cielo Epic™ platform is purpose-built to be the definitive ESG intelligence layer for enterprise operations.

Supported

Multi-framework compliance from a single data model:

G R I

Global Reporting Initiative

S A S B

Sustainability Accounting Standards

I S S B

International Sustainability Standards Board

C S R D

EU Corporate Sustainability Reporting Directive

S E C

US SEC Climate Disclosure Rules

B R S R

Business Responsibility & Sustainability Report (India)

Platform

Carbon

Scope 1, 2, and 3 emissions tracking with verifiable audit trails and automatic factor updates.

Sustainability

Auto-generated, framework-mapped sustainability reports ready for board, regulator, and investor audiences.

ESG

Real-time ESG KPI dashboards with trend analysis, peer benchmarking, and target tracking.

NeZt ero

AI-powered decarbonization roadmaps with capital allocation recommendations and milestone tracking.

Integration

The platform is designed for open integration — connecting to every major building, enterprise, hospitality, and cloud system through a comprehensive library of certified connectors and protocol adapters. No rip-and-replace required.



Building

BACnet · Modbus · OPC-UA

Native integration with all major building automation protocols — enabling seamless data acquisition from existing BMS, SCADA, and PLC infrastructure.



Hospitality

HTNG · Oracle Opera · StayNTouch

Certified integrations with leading PMS and hospitality technology platforms for full guest journey and operational data unification.



Enterprise

SAP · Oracle · Microsoft

Bi-directional integration with enterprise ERP and business intelligence platforms — placing operational intelligence in the hands of business decision-makers.



Facility

CMMS · CAFM · EAM

Native work order and asset management integration — closing the loop from predictive alert to field engineer dispatch and completion.



Cloud

AWS · Azure · Google Cloud

Multi-cloud deployment with full data portability — supporting customer cloud preferences and hybrid deployment architectures.

M o n e t i z a t i o n

The alliance operates a diversified, multi-model monetization architecture — capturing value at every layer of the stack from hardware to AI services. Recurring revenue streams dominate, creating high-quality, predictable enterprise ARR.



S a a S

Per Site · Per Room · Per Building — flexible consumption-based pricing aligned to customer footprint and expansion.



P a a S

Platform Subscription — full platform access for OEM partners, system integrators, and enterprise developers building on the ecosystem.



H a a S

Sensor Bundles — hardware-as-a-service subscription model lowering capex barriers and accelerating deployment cycles for price-sensitive markets.



D a a S

Data Services — anonymized, aggregated operational intelligence datasets licensed to research institutions, insurers, and industry benchmarking services.



A I a a S

AI Optimization — agentic AI services licensed on a consumption or outcome basis — pay-per-insight and pay-per-decision models.



E S G a a S

Compliance Services — managed ESG reporting, carbon accounting, and regulatory filing services for enterprise customers without in-house sustainability teams.



M a r k e t p l a c e

Partner Applications — revenue-share marketplace for third-party applications, vertical solutions, and data integrations built on the platform.

R e v e n u e

The alliance revenue model is structured to align incentives, reward each partner's core competency, and maximize joint market penetration — with a clear shared revenue pool for collaborative opportunities.

E n m a z

- **Instrumentation:** Hardware device sales and bundles
- **Deployment:** Site survey, installation, commissioning
- **Audits:** Energy and power quality audits
- **Consulting:** Sustainability and measurement consulting
- **Managed Services:** Ongoing device management and calibration

S h a r e d

Jointly owned and split by agreement:

- Technical Support & SLA
- Joint Project Delivery
- Recurring Platform + Hardware Bundles



Recurring revenue shared pool creates long-term alignment and predictable joint ARR growth.

I n s p i r o n i c s

- **Platforms:** CK, CE, CX, CD, CM SaaS subscriptions
- **Digital Twin:** Twin modeling, deployment, and licensing
- **AI Services:** Agentic AI and ML model subscriptions
- **ESG:** Compliance reporting and net-zero planning
- **Analytics:** Advanced analytics and insights services
- **IP:** Platform IP licensing to OEM and white-label partners

C o m p e t i t i v e

The Inspironics + Enmaz alliance competes in a landscape dominated by large industrial incumbents — but wins on agility, openness, and native AI-ESG integration that legacy players cannot match without multi-year re-architecture programs.

P r i m a r y

- **Siemens** — Building X, MindSphere
- **Schneider Electric** — EcoStruxure, EcoStruxure IT
- **Honeywell** — Forge, Connected Buildings
- **Johnson Controls** — OpenBlue
- **ABB** — ABB Ability

Common incumbent weakness: **closed ecosystems, legacy tech debt, slow ESG/AI roadmaps, and high integration costs.**

A l l i a n c e

O p e n

No vendor lock-in; integrates with any sensor, protocol, or cloud

D i g i t a l

Twin-first architecture, not a bolt-on feature

E S G

Multi-framework ESG built into the core data model

A g e n t i c

Live autonomous AI agents — not just dashboards

H o s p i t a l i t y

Unique cross-vertical expertise in a single platform

M u l t i t h o u d

AWS, Azure, Google — customer choice, not vendor choice

P a r t n e r

Open marketplace and OEM/white-label programs

Strategic

The alliance's long-term defensibility is built on two reinforcing layers of competitive moat — technical assets that are difficult to replicate, and business model advantages that compound over time.

Technical

Hard-to-replicate technology assets that create durable competitive protection:

- **Patents:** Sensor design, AI agent architecture, and digital twin methodologies
- **Digital Twin Models:** Pre-built, calibrated vertical twin templates (Hospitality, Healthcare, Manufacturing, Data Center)
- **AI Agents:** Trained, production-grade agentic AI models with operational history
- **Data Fabric:** Proprietary enterprise data model and integration framework
- **Industry Templates:** Pre-configured vertical solutions reducing deployment time by 60%+

Business

Structural advantages that grow stronger with every customer and partner added:

- **Partner Ecosystem:** OEM, SI, and distribution partnerships creating multi-channel, low-CAC growth
- **Recurring Revenue:** High-retention SaaS and PaaS subscriptions generating compounding ARR
- **Cross-Industry Reusability:** Core platform investments amortized across 6+ verticals
- **Consulting + Platform Model:** Advisory relationships that convert to long-term platform commitments

☐ The combination of IP-protected technology assets with a network-effect-driven partner ecosystem creates a **reinforcing moat** — one that widens with each deployment, partner, and data point added to the system.

R O I

The platform delivers quantifiable, auditable returns across six performance dimensions — with a typical payback period of **6–18 months**, making enterprise investment decisions straightforward for CFOs and procurement teams.

1 –3

E n e r g y

Reduction in energy consumption through AI-driven HVAC, lighting, and utility optimization

2 –4

M a i n t e n a n c e

Reduction in unplanned maintenance and emergency repair costs through predictive maintenance

1 –2

C a r b o n

Decrease in Scope 1 and 2 carbon emissions supporting net-zero commitments

1 –2

O c c u p a n c y

Improvement in space utilization and occupant productivity through intelligent space management

1 –3

A s s e t

Increase in asset operational lifespan through condition-based maintenance and predictive analytics

6–1

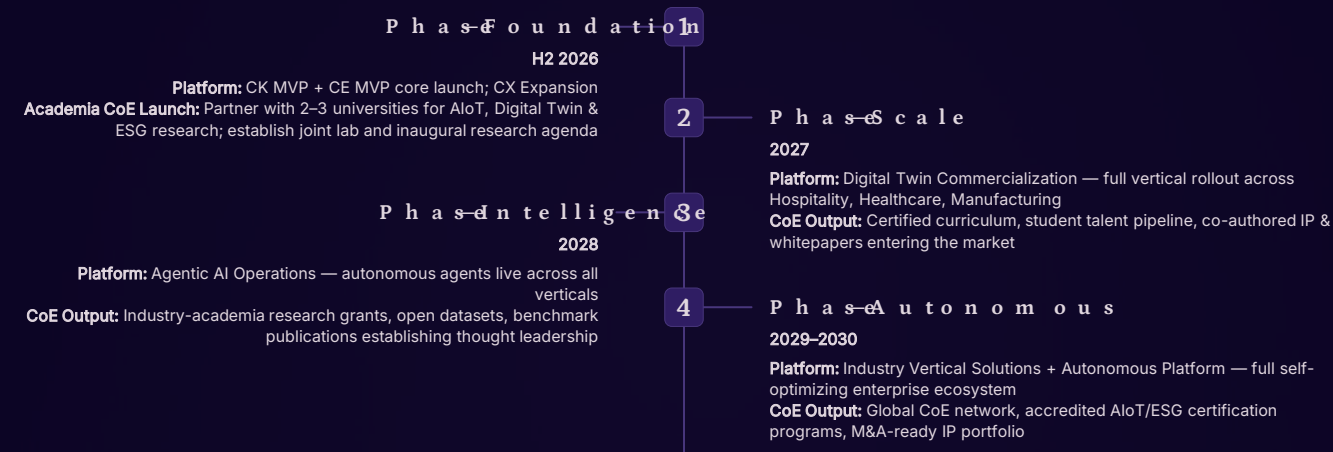
T y p i c a l

Time to positive ROI from initial deployment — driven by energy and maintenance savings

🕒 All ROI targets are based on production deployments and industry benchmarks. The CE — Cielo Epic[®] platform generates the audit-ready evidence trail required for board-level ROI reporting and ESG disclosure.

C o m p r e s s e d

Four focused phases from platform foundation to fully autonomous enterprise — with Academia Centers of Excellence (CoE) embedded as a **cross-cutting enabler** at every phase, building talent pipeline, open IP, and global credibility in parallel with commercial milestones.



The **Academia CoE is not a standalone initiative** — it is a parallel enabler track that produces talent, validated IP, and credibility assets that directly accelerate commercial platform adoption at each phase milestone.

U n i v e r s i t y

2–3 founding university partners in AIoT, Digital Twin & ESG research; joint lab infrastructure established in Phase 1

C e r t i f i e d

Industry-recognized AIoT and ESG certification programs co-developed with academic partners; first cohort in Phase 2

R e s e a r c h

Competitive research grants and co-authored publications in Phase 3; open datasets strengthening platform credibility

G l o b a l

Multi-geography CoE network by Phase 4; accredited programs and M&A-ready IP portfolio supporting enterprise and investor narratives